

MRC Lecture 3 — Derivation Guide 2

The Symmetric Loss and Why Other Contrastive Losses Fail

Sai T. Reddy — ETH Zurich / BIIE

1. Motivation

In Derivation Guide 1, we derived the InfoNCE loss for the **agent**→**target** direction: for each agent, maximize the probability of selecting the correct target from the competitor pool. But molecular binding is symmetric — the free energy of the antibody-antigen complex is the same whether computed from the antibody’s perspective or the antigen’s perspective: $\Delta G_{AB} = \Delta G_{BA}$. This physical symmetry requires that the training loss be bidirectional.

This guide derives the symmetric loss from binding symmetry, then shows why three alternative contrastive losses — triplet loss, binary cross-entropy, and margin-based loss — each violate at least one of the physical axioms that the convergence equation satisfies.

2. The Symmetric Loss

2.1 The Two Directions

Agent→**target (row-wise)**. For each agent i , compute the probability of selecting the correct target from the batch. This is the InfoNCE loss from Derivation Guide 1:

$$L_{\text{agent} \rightarrow \text{target}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(S_{ii})}{\sum_{j=1}^N \exp(S_{ij})}$$

This applies softmax **row-wise** to the similarity matrix S . Each row becomes a probability distribution: “for agent i , which target?”

Target→**agent (column-wise)**. For each target j , compute the probability of being selected by the correct agent from the batch:

$$L_{\text{target} \rightarrow \text{agent}} = -\frac{1}{N} \sum_{j=1}^N \log \frac{\exp(S_{jj})}{\sum_{i=1}^N \exp(S_{ij})}$$

This applies softmax **column-wise** to the similarity matrix S . Each column becomes a probability distribution: “for target j , which agent?”

2.2 The Symmetric Loss

The symmetric loss averages both directions:

$$L_{\text{symmetric}} = \frac{1}{2} (L_{\text{agent} \rightarrow \text{target}} + L_{\text{target} \rightarrow \text{agent}})$$

2.3 Physical Derivation

Claim. The symmetric loss is required by the physical symmetry of binding free energy.

Argument. The binding free energy is a property of the molecular complex, not of either partner individually:

$$\Delta G_{AB} = \Delta G_{BA}$$

This means the embedding space must satisfy: $\text{sim}(q_A, k_B) = \text{sim}(q_B, k_A)$ when the embeddings correctly represent binding energies. Training with only the agent→target loss does not enforce this

— it optimizes the embedding from the agent’s perspective but imposes no constraint on the reverse direction. The target→agent loss provides the complementary constraint.

CLIP (Radford et al., 2021) introduced the symmetric loss empirically — it produced better-balanced embedding spaces. The MRC framework derives it from physics: binding symmetry requires bidirectional training.

2.4 Practical Consequence

CALM (Lee et al., 2026) confirmed that the symmetric loss produces balanced retrieval performance: the Ag→Ab and Ab→Ag retrieval metrics have overlapping confidence intervals, validating that the symmetric loss enforces a balanced embedding space where neither molecular modality dominates.

3. Why Triplet Loss Fails

Definition. The triplet loss (Schroff et al., *CVPR*, 2015) takes an anchor a , a positive p , and a single negative n , and enforces a margin:

$$L_{\text{triplet}} = \max(0, \text{sim}(a, n) - \text{sim}(a, p) + \alpha)$$

where $\alpha > 0$ is a fixed margin hyperparameter.

What it does. The loss requires that the positive is more similar to the anchor than the negative, by at least a margin α . If the margin is satisfied, the loss is zero and no gradient flows.

Which axiom it violates: Normalization.

The Boltzmann distribution assigns probability based on the scores of **all** competitors simultaneously — the denominator is the sum over the entire pool:

$$P(f) = \frac{\exp(s_f)}{\exp(s_f) + \exp(s_{c_1}) + \dots + \exp(s_{c_M})}$$

Triplet loss considers only **one** negative at a time. It does not model the full competitor pool. The probability implied by triplet loss is not a valid probability distribution over all candidates — it does not normalize over the full set of alternatives.

Consequence. Triplet loss can push the positive closer and the negative farther, but it cannot learn the correct ranking over a pool of many competitors. An antibody encounters thousands of potential antigens simultaneously, not one at a time. A loss function that considers only one negative discards information about the competitive landscape.

Additional problem. The margin α is a hyperparameter with no physical interpretation in the convergence equation. The convergence equation’s selection probability is determined entirely by the scores and the temperature — there is no margin parameter.

4. Why Binary Cross-Entropy Fails

Definition. Binary cross-entropy (BCE) treats each agent-target pair independently as a binary classification (binding vs. non-binding):

$$L_{\text{BCE}} = -y \log \sigma(s) - (1 - y) \log (1 - \sigma(s))$$

where $y \in \{0, 1\}$ is the binary label and $\sigma(s) = 1 / (1 + \exp(-s))$ is the sigmoid function.

What it does. For each pair, the model predicts the probability of binding using the sigmoid of the similarity score. Positive pairs ($y = 1$) should have high sigmoid output; negative pairs ($y = 0$) should have low sigmoid output.

Which axiom it violates: Independence of Irrelevant Alternatives (IIA).

The sigmoid function was shown to violate IIA in Lecture 1 (Derivation Guide 5, preview) and will be proven formally in Lecture 6. The ratio of sigmoid outputs for two scores s_A and s_B is:

$$\frac{\sigma(s_A)}{\sigma(s_B)} = \frac{1 + \exp(-s_B)}{1 + \exp(-s_A)}$$

This ratio depends on the **absolute values** of s_A and s_B , not just their difference. The relative preference between two targets changes depending on the absolute score scale — which violates IIA. **Consequence.** BCE models **absolute** binding probability (does this pair bind?), not **relative** selection among competitors (which target does this agent select from a pool?). But molecular recognition is inherently competitive — an antibody’s binding to one antigen depends on what other antigens are available. BCE cannot capture this competitive structure.

Additional problem. BCE treats each pair independently. In a batch of N pairs, there are N positive pairs and $N(N-1)$ negative pairs, but BCE processes each independently without modeling the joint distribution. InfoNCE models the full $N \times N$ similarity matrix and computes selection probabilities over the entire pool.

5. Why Margin-Based Contrastive Loss Fails

Definition. The margin-based contrastive loss (Hadsell et al., *CVPR*, 2006) uses a distance threshold:

$$L_{\text{margin}} = y \cdot d^2 + (1 - y) \cdot \max(0, m - d)^2$$

where d is the distance between embeddings, y is the label, and m is the margin.

What it does. For positive pairs ($y = 1$): minimize the distance. For negative pairs ($y = 0$): push the distance beyond a margin m . Once the negative is beyond the margin, no further gradient is applied.

Which axiom it violates: Positivity (soft selection).

The Boltzmann distribution assigns **non-zero probability** to every candidate (Axiom 1: positivity). At non-zero temperature, even energetically unfavorable binding events have non-zero probability. The margin-based loss imposes a **hard boundary**: once a negative is pushed beyond the margin, it is effectively assigned zero probability. This contradicts the soft, probabilistic selection required by thermodynamics.

Consequence. The margin-based loss produces an embedding space with hard boundaries between binding and non-binding regions. The Boltzmann distribution produces soft, graded probabilities that decrease smoothly with decreasing compatibility. The margin-based geometry does not match the physics.

Additional problems. Like triplet loss, the margin m is a hyperparameter with no physical interpretation. Like BCE, the loss processes pairs independently without modeling the full competitor pool.

6. Summary Comparison

Loss function	Models full pool?	Selection function	Physical axiom violated	Has unphysical hyperparameters?
InfoNCE	Yes (all in-batch)	Softmax (convergence eq.)	None	No (τ is physical)
Triplet	No (one negative)	Margin comparison	Normalization	Yes (margin α)
Binary CE	No (pairwise)	Sigmoid	IIA	No
Margin-based	No (pairwise)	Hard threshold	Positivity (soft selection)	Yes (margin m)

Conclusion. InfoNCE is the **only** contrastive loss that:

1. Models the full competitor pool (via the softmax denominator)
2. Uses the convergence equation as the selection function
3. Violates none of the five physical axioms
4. Has no hyperparameters beyond the physically meaningful temperature τ

Training with InfoNCE trains the model to reproduce the Boltzmann selection probability. Training with any other objective trains the model to reproduce a different, non-physical target.

7. Checkpoint Questions

Q1. A researcher argues: “Triplet loss is simpler and faster to compute than InfoNCE. For a large enough batch with many triplets, the results should converge.” Is this correct?

Answer: Even with many triplets, triplet loss never models the full competitor pool simultaneously. Each triplet considers one positive and one negative — it optimizes pairwise relationships, not the joint distribution over all candidates. InfoNCE computes a softmax over the entire batch, modeling the competitive selection that molecular recognition requires. The approximation gap does not close with more triplets because the loss function’s structure is fundamentally different from the Boltzmann selection probability.

Q2. The symmetric loss averages the row-wise and column-wise InfoNCE. Could you use only one direction?

Answer: You could, and the model would still learn useful embeddings. But the embedding space would be unbalanced: the agent modality would be better discriminated than the target modality (or vice versa). The physical requirement $\Delta G_{AB} = \Delta G_{BA}$ means both directions should be equally well-learned. CALM confirmed this empirically: the symmetric loss produces overlapping Ag→Ab and Ab→Ag retrieval metrics, while asymmetric loss produces a gap.

Q3. Binary cross-entropy uses the sigmoid function. InfoNCE uses the softmax function. Both are “smooth” and “differentiable.” Why does the choice matter?

Answer: Smoothness and differentiability are properties needed for gradient-based optimization — both losses can be trained with backpropagation. The difference is what function the model learns to reproduce. Sigmoid models absolute binding probability independently for each pair. Softmax models relative selection probability over a pool. Molecular recognition is competitive — whether an antibody binds antigen A depends on what other antigens are available. Softmax captures this competition; sigmoid does not. The mathematical distinction is IIA: softmax satisfies it (probability ratios depend only on score differences); sigmoid does not (ratios depend on absolute scores).

8. Connection to Future Lectures

- **Lecture 6:** The five axioms of competitive selection are stated formally and the uniqueness of softmax is proven. The elimination of sigmoid (IIA violation) is derived on the whiteboard. This provides the theoretical foundation for why BCE fails.
- **Lecture 7:** The SFM architecture prescribes InfoNCE as the training objective and the symmetric loss as a consequence of binding energy symmetry. Both are listed in the “prescribed by theory” column of the architecture table.
- **Lecture 9:** The MRC scaling law depends on InfoNCE training — it predicts exponential convergence because InfoNCE is the MLE of the correct governing equation. If a different loss were used, the scaling prediction would not apply.